

Social and cognitive drivers of generative AI adoption: A unified socio-cognitive model for engineering education

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ABSTRACT

Generative artificial intelligence (AI) redefines cognitive and social processes in learning environments, yet its psychological and contextual determinants in developing countries remain underexplored. This study investigates engineering students' adoption of generative AI by integrating the Technology Acceptance Model (TAM) with the Unified Theory of Acceptance and Use of Technology (UTAUT) to develop a unified socio-cognitive framework. Survey data from 378 Bangladeshi engineering students were analysed using PLS-SEM and multi-group comparisons to evaluate structural, mediating, and demographic effects. Results demonstrate that behavioural intention is the strongest predictor of AI use, with perceived usefulness and ease of use functioning as key mediators linking cognitive, social, and attitudinal factors. Job relevance, result demonstrability, and subjective norms exert significant direct and indirect influences, revealing a multilayered decision-making process. The model explains 64% of the variance in usage behaviour, affirming the robustness of this integrated framework. Multi-group analysis (MGA) results indicate that domestic students perceive a higher level of perceived ease of use than their international peers, while the pattern is reversed in terms of students' image when interacting with generative AI tools. Gender shows no significant differences for either group. The study contributes to human–technology interaction research by clarifying the cognitive and social drivers underlying generative AI acceptance. It emphasises the role of contextual constraints and user perceptions in shaping technology-driven learning behaviour. Theoretically, the study extends AI acceptance literature by contextualising model validation within a resource-constrained, multicultural higher education environment. Practically, it offers evidence-based guidance for designing culturally responsive and cognitively supportive AI learning ecosystems. The findings underscore the need for cross-cultural, longitudinal research to unpack evolving patterns of AI adoption in education.

1. Introduction

Over the past decade, generative artificial intelligence (GenAI) has transformed how humans interact with knowledge systems. Enabled by rapid advances in machine learning and large language models, tools such as ChatGPT, Gemini, and DeepSeek have moved beyond predefined automation to context-sensitive reasoning and content generation (Asag et al., 2024, pp. 483–489; Nay, 2022). Their diffusion into higher education signals a profound pedagogical shift such that GenAI now supports personalised tutoring, automated feedback, and creative knowledge construction (Baidoo-Anu & Ansah, 2023; Dahri et al., 2024). While these technologies promise to augment human learning, their effective adoption depends not merely on technical affordances but on users' socio-cognitive evaluations, such as trust, perceived

usefulness, and social influence, that determine sustained engagement (Shahzad et al., 2024).

Technology acceptance research, particularly the Technology Acceptance Model (TAM) (Davis, 1989) and the Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh & Davis, 2003), provides robust frameworks for explaining such behavioural intentions. These models emphasise perceived usefulness (PU) and perceived ease of use (PEOU) as core cognitive antecedents, while integrating social influence, image, and facilitating conditions as key contextual factors. Prior research has consistently shown that integrating multiple technology acceptance frameworks provides a more robust explanation of educational technology adoption than relying on a single theoretical lens. Early foundational work combining the Technology Acceptance Model (TAM) with social and contextual theories, such as the Theory of

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Planned Behaviour (TPB) and Innovation Diffusion Theory (IDT), demonstrated that perceived usefulness and ease of use alone were insufficient to explain learners' behavioural intentions without accounting for subjective norms, facilitating conditions, and institutional context (Ajzen, 1991; Rogers et al., 2008; Venkatesh & Davis, 2000). These early integrations established the theoretical rationale for viewing technology adoption as a socio-cognitive process rather than an internally driven decision-making process. Moreover, the applications of TAM and UTAUT to investigate educational AI adoption are predominantly situated in developed, resource-rich contexts, leaving a limited understanding of how social and cognitive forces interact in low-resource, collectivist learning environments (Ruiz-Rojas et al., 2024; Tiwari et al., 2024).

Building on this foundation, UTAUT and UTAUT2 further formalised the integration of cognitive beliefs, social influence, and environmental enablers, and were widely validated in educational settings (Venkatesh et al., 2003). Subsequent studies extended these models by incorporating motivational, pedagogical, and affective constructs, such as self-efficacy, perceived enjoyment, trust, and learning value, when examining the adoption of learning management systems, mobile learning, MOOCs, and virtual learning environments (Teo, 2011; Šumak et al., 2011). Meta-analytical evidence also confirms that extended or integrated acceptance models yield significantly higher explanatory power for behavioural intention and usage behaviour in education compared to baseline TAM (King & He, 2006).

More recent research has applied integrated TAM-UTAUT and socio-technical models to emerging educational technologies, including mobile learning, learning analytics dashboards, AI-based recommender systems, and intelligent tutoring systems (Dwivedi et al., 2019; Sitar-Taut & Mican, 2021). These studies consistently demonstrate that learners' adoption decisions are shaped by interacting cognitive evaluations (e.g., usefulness, effort expectancy), social pressures (e.g., peer and instructor influence), and structural conditions (e.g., institutional support, digital infrastructure). Importantly, they also highlight that educational technologies embedded in complex pedagogical practices require integrative models that capture both individual agency and contextual constraints.

Despite this growing body of work, most integrated acceptance studies remain situated in non-generative AI contexts, focus on general student populations, and are predominantly conducted in developed or resource-rich educational environments. Engineering education, characterised by problem-solving intensity, disciplinary epistemologies, and high cognitive load, has been comparatively underrepresented in technology acceptance research, particularly with respect to generative AI tools that actively co-produce knowledge rather than merely deliver content. As a result, it remains empirically unclear whether established integrated socio-cognitive models adequately capture the adoption dynamics of generative AI in engineering education, especially in developing-country and multicultural contexts, where infrastructural, cultural, and pedagogical conditions may differ substantially.

In developing countries such as Bangladesh, where digital infrastructure and institutional support remain uneven, the adoption of GenAI presents unique challenges. Limited access to technology, inadequate faculty expertise, and weak policy frameworks often constrain equitable use (Awal & Haque, 2025; Borg & Östergren, 2015). Beyond these structural barriers, social norms and peer networks strongly shape students' attitudes and behaviours towards technology. In collectivist educational cultures, where conformity and group approval are highly valued, peer influence may either amplify or suppress the perceived utility of GenAI tools (Sobaih et al., 2024). Yet, little is known about how these social cues interact with cognitive perceptions such as PU, PEOU, job relevance, and result demonstrability to influence students' intentions and actual use of GenAI, especially within engineering education, where domain-specific applications like coding and simulation are prevalent (Asag et al., 2024, pp. 483–489).

While these structural barriers form an essential backdrop to

generative AI adoption in developing countries, the present study does not model them as direct predictors. Instead, they are conceptualised as contextual boundary conditions that shape how students evaluate and engage with AI technologies. In resource-constrained environments, limitations in infrastructure, institutional support, and policy frameworks are often internalised through users' perceptions of usefulness, ease of use, and demonstrability, as well as through socially mediated learning practices. Accordingly, this study focuses on proximal socio-cognitive determinants while analytically embedding structural constraints within the broader contextual environment influencing these perceptions.

Guided by the integrated UTAUT-TAM2 framework, this study aims to investigate how social cues and cognitive evaluations combinedly influence engineering students' behavioural intentions and actual use of generative AI tools in a low-resource, cross-cultural university setting in Bangladesh. The integrated framework draws selectively on constructs that have been theoretically and empirically validated as core determinants of technology adoption in educational settings. Following TAM2, behavioural intention to use GenAI is shaped by key social and cognitive determinants such as subjective norm, image, job relevance, and perceived usefulness (Venkatesh & Davis, 2000). Specifically, it examines how subjective norms and student image affect perceived usefulness and perceived ease of use and how job relevance, output quality, and result demonstrability contribute to perceived usefulness, ease of use, and intention. By analysing behavioural differences across gender and domestic–international student groups, this research further offers a contextually grounded understanding of AI adoption in under-resourced educational settings.

While the integrated UTAUT-TAM2 framework captures general socio-cognitive mechanisms underlying technology adoption, this study applies and interprets the model within the specific context of engineering education. Engineering students' engagement with generative AI is shaped by discipline-specific tasks, including coding, simulation, and design problem-solving, which influence how constructs such as job relevance, output quality, and result demonstrability are perceived. Accordingly, the model is theoretically transferable across academic disciplines; however, its empirical instantiation and interpretation are context dependent. The study advances AI adoption research by extending the UTAUT-TAM integration to a Global South context, empirically linking social and cognitive constructs that have rarely been examined together, and offering practical insights for culturally responsive and inclusive AI integration in higher education.

2. Transforming learning and GenAI adoption: social and cognitive dynamics

In engineering education, artificial intelligence has redefined how students learn, collaborate, and conceptualise complex problems by providing intelligent feedback systems, adaptive simulations, and real-time diagnostic support (Ruiz-Rojas et al., 2024). Building on these affordances, generative AI tools, capable of producing human-like text, code, and visuals, introduce a new form of cognitive partnership in which students engage with algorithms not only as computational aids but also as co-creators of ideas (Acosta-Enriquez et al., 2024; Asag et al., 2024, pp. 483–489). These tools expand the pedagogical horizon beyond efficiency and accuracy, stimulate divergent thinking, enhance creative reasoning, and provide iterative feedback loops that mirror authentic engineering design processes (Al-kfairy, 2024). Such developments signal a pedagogical transformation from instructor-centred to learner-augmented knowledge creation.

Despite the clear educational promise, what fundamentally drives GenAI adoption remains contested. Empirical findings diverge on whether social or cognitive factors play the dominant role. Studies grounded in UTAUT and related frameworks identify social influence, peer norms, and image as critical determinants, particularly in collectivist cultures where conformity and reputation shape digital behaviours

(Sobaih et al., 2024; Yilmaz et al., 2024; Zhao et al., 2024). Conversely, others highlight that adoption is primarily cognitively motivated, anchored in perceptions of usefulness, ease of use, and performance enhancement, suggesting that even in socially dense environments, rational evaluation may override normative pressure (Al-Abdullatif, 2024; Almogren et al., 2024). This theoretical divide reflects a broader tension between social diffusion logics (adoption as socially contagious) and cognitive evaluation models (adoption as a rational choice).

However, these competing explanations are not mutually exclusive but context dependent. The strength and direction of social and cognitive effects often vary across institutional cultures, infrastructural capacities, and national technology policies (Bouteraa et al., 2024; Ruiz-Rojas et al., 2024). For instance, in well-resourced contexts, individual perceptions of task fit, or tool usability, may dominate, whereas in low-resource environments, adoption may depend more on social scaffolding, peer learning, and informal support systems. Engineering education further complicates this relationship. Students' high task orientation and reliance on peer validation can make their adoption decisions simultaneously rational and relational.

While concerns such as algorithmic bias, data privacy, and persistent scepticism among educators are critical to understanding the broader discourse surrounding generative AI in education (Sallam et al., 2024), the present study does not model these issues as direct predictors of adoption. Instead, they are treated as contextual and normative conditions that shape how students evaluate the legitimacy, usefulness, and social acceptability of generative AI tools. In line with socio-cognitive acceptance theory, the influence of such concerns is expected to manifest indirectly through students' cognitive evaluations (e.g., performance expectancy and output quality) and social influence mechanisms (e.g., subjective norm), rather than as isolated determinants.

In the Global South, the adoption of generative AI (GenAI) is still in its formative stage. Research in these regions remains limited, despite distinct socio-economic and institutional realities that shape how technology diffuses across higher education systems (Adams et al., 2025; Henadirage & Gunarathne, 2025; Jin et al., 2025). Particularly in Bangladesh, the adoption of generative AI (GenAI) remains emergent and uneven. While infrastructural limitations, policy gaps, and resource constraints continue to shape technology diffusion (Anik & Rahman, 2025), adoption is increasingly propelled by bottom-up, socially driven practices rather than institutional initiatives (Asag et al., 2024, pp. 483–489). Studies reveal that Bangladeshi students often rely on peer networks, curiosity, and experimentation to integrate tools like ChatGPT into academic routines, compensating for the absence of structured AI training or institutional guidance (Anik & Rahman, 2025; Saidu & Mamun, 2025). This informal, peer-mediated diffusion underscores a culturally grounded mode of adoption shaped by trust, ethics, and emotional regulation that extends beyond the traditional cognitive determinants of perceived usefulness and ease of use (Asag et al., 2024, pp. 483–489; Saidu & Al Mamun, 2026). At the same time, persistent scepticism among educators regarding academic integrity and ethical risks (Jaiteh et al., 2024; Saidu & Mamun, 2025) reinforces the tension between institutional caution and student-led innovation.

Despite growing optimism, recent research highlights concerns about overreliance on GenAI tools and the potential erosion of students' critical thinking (Foroughi et al., 2024). While Bangladeshi students frequently use ChatGPT for paraphrasing or drafting assignments (Jaiteh et al., 2024), recent study argues that dependence may undermine originality (Saidu & Mamun, 2025). Similarly, issues of accuracy, bias, and data privacy persist as major deterrents in other social contexts (Dahri et al., 2024). These findings suggest that perceived usefulness and ease of use alone may not ensure sustained adoption; rather, factors such as trust, ethical awareness, and self-regulation increasingly mediate the relationship between intention and behaviour.

Taken together, these dynamics suggest that GenAI adoption in Bangladesh is not solely a technological process but a socially and emotionally embedded phenomenon, where collective norms and

contextual constraints jointly mediate the relationship between intention and use. Consequently, the literature indicates that understanding GenAI adoption requires moving beyond a binary distinction between social and cognitive determinants. Instead, these pathways interact dynamically: social cues shape perceptions of usefulness, while cognitive appraisals reinforce social legitimacy. This interplay is particularly salient in Global South contexts, where institutional and infrastructural conditions differ markedly from those in the Global North. Overall, the literature reveals both convergence and contradiction regarding the social and cognitive mechanisms of GenAI adoption in higher education. Perceived usefulness and ease of use consistently emerge as central predictors, yet the strength of social influence and contextual constraints varies across settings. To address these variances, the present study adopts an integrated UTAUT–TAM2 framework to examine how cognitive evaluations and social influence mechanisms jointly shape engineering students' adoption of generative AI in Bangladesh, while accounting for the broader ethical, institutional, and normative context in which such technologies are embedded.

3. Theoretical framework and research model

Building a comprehensive understanding of students' adoption of generative AI requires integrating insights from multiple theoretical traditions that illuminate both cognitive evaluation and social influence processes. The Value-Based Adoption Model (Kim et al., 2007) emphasises the trade-off between perceived benefits and risks, critical for understanding students' cautious adoption of ChatGPT amid integrity concerns (Al-Abdullatif, 2024). The Diffusion of Innovation theory (Rogers et al., 2008) explains how perceived relative advantage, compatibility, and observability affect adoption, though it is less predictive of individual behavioural intentions (Al-kfairy, 2024). Social Cognitive Theory (Bandura, 1977, 1986) and the Theory of Planned Behaviour (Ajzen, 1991) further illuminate the social and behavioural processes underpinning AI adoption. In collectivist learning environments, subjective norms and perceived behavioural control often outweigh individual attitudes, suggesting that social cues are powerful yet understudied determinants of GenAI adoption (Ruiz-Rojas et al., 2024). Recent empirical findings also point to image and peer endorsement as significant motivators, as students increasingly view AI competency as a symbol of academic modernity (Asag et al., 2024, pp. 483–489).

Despite the diversity of theoretical frameworks, two overarching themes dominate current literature. First, cognitive appraisal processes, how individuals evaluate the usefulness, effort, and relevance of emerging technologies, remain central to explaining adoption behaviour (Foroughi et al., 2024; Zhao et al., 2024). Second, social and cultural dynamics, including collective norms, peer expectations, and institutional support, show inconsistent influence across contexts, suggesting a limited understanding of how culturally embedded values shape these cognitive judgments (Sobaih et al., 2024; Tiwari et al., 2024). Moreover, existing research is disproportionately concentrated in technologically advanced and resource-rich environments, leaving underexplored the realities of low-resource settings where infrastructural, linguistic, and policy constraints critically mediate technology acceptance (Awal & Haque, 2025). This study excludes the collective norms, peer expectations, and institutional support in developing the research model. It was a deliberate methodological decision intended to preserve model parsimony and focus on the primary cognitive-functional drivers of technology adoption. Theoretically, these factors are often partially captured within the *Subjective Norms* construct; introducing them as separate variables risked significant multicollinearity, which would have obscured the statistical clarity of the results. Furthermore, in the specific context of a resource-limited environment, institutional support structures are frequently in a nascent or inconsistent stage, making their inclusion less likely to yield statistically significant or generalizable insights compared to internal cognitive appraisals.

Drawing upon these insights, the present study advances an integrated UTAUT–TAM2 framework that conceptually fuses the social-influence mechanisms of UTAUT with the cognitive appraisal dimensions of TAM2. The original Technology Acceptance Model (TAM) (Davis, 1989) has served as a cornerstone in explaining technology adoption, positing perceived usefulness (PU) and perceived ease of use (PEOU) as key cognitive determinants of behavioural intention (Al-Abdullatif, 2024; Foroughi et al., 2024; Tiwari et al., 2024). Despite its enduring relevance, the original TAM has been critiqued for its limited attention to social and contextual influences that substantially shape user behaviour (Mogaji et al., 2024). To overcome these limitations, subsequent extensions such as the Technology Acceptance Model 2 (TAM2) (Venkatesh & Davis, 2000) integrated social influence processes, including Subjective Norm (SN) and Student's Image (SI), alongside cognitive instrumental factors such as Job Relevance (JR), Output Quality (OQ), and Result Demonstrability (RD). This expanded framework provides a more comprehensive explanation of adoption behaviours by recognising that individuals' technology acceptance is influenced not only by cognitive appraisals but also by perceived social expectations and task-related utility (Almogren et al., 2024; Tiwari et al., 2024). As such, TAM2 offers a theoretically grounded lens for understanding the nuanced interplay of cognitive and social determinants in students' adoption of generative AI technologies. Furthermore, cognitive instrumental factors derived from TAM2, such as job relevance, output quality, and result demonstrability, have gained renewed importance in GenAI-supported education. Empirical studies report that students critically evaluate the contextual relevance, accuracy, and explainability of AI-generated outputs before integrating them into academic tasks, particularly in technical and engineering domains where correctness and traceability are essential (Bauer et al., 2021; Deng & Lin, 2022). The visibility of AI-generated results and their perceived contribution to learning outcomes also shape continued use intentions, aligning closely with the notion of result demonstrability in the Technology Acceptance Model (TAM2).

Building on TAM and its extensions, the UTAUT (Venkatesh et al., 2003) offers one of the most comprehensive and empirically validated models for explaining technology adoption behaviours. It integrates constructs from multiple earlier frameworks, including the TAM and Theory of Planned Behaviour (TPB), to explain users' behavioural intentions and actual technology use. It integrates four key constructs, such as performance expectancy, effort expectancy, social influence, and facilitating conditions, to conceptualise technology adoption as a function of interacting cognitive and social drivers. This multidimensional structure has proven especially relevant in educational contexts, where technology use is shaped by both individual perceptions and collective learning environments (Almogren et al., 2024; Sobaih et al., 2024). Within this tradition, performance expectancy represents users' instrumental evaluations of the system's effectiveness and usefulness (Venkatesh et al., 2003), while effort expectancy captures the degree of ease associated with the use of the system (Venkatesh & Davis, 2000). These constructs are especially salient in learning environments where effort expectancy and performance outcomes are central, such as engineering education. More recent research suggests that these foundational acceptance mechanisms remain highly relevant in the context of Generative AI-enabled learning technologies, including large language models and AI-assisted tutoring systems. Studies examining student adoption of GenAI tools (e.g., ChatGPT, Copilot, and AI-based content generators) indicate that performance expectancy continues to be a dominant predictor of intention, particularly when learners perceive AI systems as enhancing productivity, problem-solving efficiency, and academic performance (Baidoo-Anu & Ansah, 2023; Karakas, 2023). Similarly, effort expectancy has been shown to influence students' willingness to engage with GenAI tools, as intuitive interaction, natural language interfaces, and reduced cognitive effort lower barriers to adoption (Tlili et al., 2025; Zhai, 2024).

Social influence is theorised in TAM2 and UTAUT as a normative

process shaped by perceived expectations of significant others and the social value attached to system use. Accordingly, Subjective Norm and Image have been consistently employed to represent normative pressure and reputational considerations in educational technology adoption studies (Dwivedi et al., 2019; Venkatesh & Davis, 2000). Together, these constructs provide a theoretically coherent representation of socio-cognitive influences on technology use intentions. Social influence mechanisms likewise appear amplified in the GenAI context. Recent studies show that institutional policies, instructors' attitudes, peer usage norms, and broader academic discourse around ethical and responsible AI use significantly shape students' adoption decisions (Chan & Hu, 2023; Farrokhnia et al., 2024). In this regard, Subjective Norm captures perceived expectations from instructors and peers regarding acceptable AI use, while Image reflects concerns about academic reputation, legitimacy, and perceptions of competence when using GenAI tools. These findings reinforce prior assertions that normative and reputational factors are central to technology adoption in educational settings, particularly for emerging and socially contested technologies such as Generative AI. Although constructs such as trust, ethical awareness, and self-regulation are relevant to generative AI use in educational settings, they represent distinct theoretical dimensions rather than components of social influence. While trust concerns system reliability, and ethical awareness involves moral reasoning, self-regulation pertains to internal metacognitive control. Grouping these under 'social influence' would conflate external normative pressures with internal cognitive and moral processes. To preserve theoretical coherence and model parsimony, the study, therefore, focuses on a core set of socio-cognitive constructs central to the integrated UTAUT–TAM2 framework.

Table 1 summarises how TAM2's detailed cognitive and social constructs are conceptually embedded within UTAUT's higher-order determinants, forming an integrated socio-cognitive framework for analysing GenAI adoption. At the macro level, the UTAUT identifies four higher-order determinants of technology adoption, e.g., performance expectancy, effort expectancy, social influence, and facilitating conditions that synthesise earlier acceptance models, in contrast to the more granular cognitive emphasis of TAM2. TAM2, in contrast, elaborates these determinants through more specific cognitive instrumental factors

Table 1
Mapping of TAM2 constructs with the UTAUT framework.

UTAUT Core Determinants	Corresponding TAM2 Constructs Under Investigation	Conceptual Meaning in This Study
Performance Expectancy	Perceived Usefulness (PU), Job Relevance (JR), Output Quality (OQ), Result Demonstrability (RD)	The degree to which students believe GenAI tools enhance learning performance, relevance to coursework, output accuracy, and demonstrable academic results.
Effort Expectancy	Perceived Ease of Use (PEOU)	The extent to which students perceive GenAI tools as intuitive and requiring minimal effort to learn and use effectively.
Social Influence	Subjective Norm (SN), Student Image (SI)	The perceived social pressure from peers and instructors to use GenAI, and the belief that using it enhances one's academic reputation or social standing.
Facilitating Conditions	Contextual Institutional Supports	Availability of institutional infrastructure, policies, and peer networks that enable or constrain effective GenAI adoption.

Note: Table 1 summarises theoretically relevant constructs; only those aligned with the study's parsimonious UTAUT–TAM2 framework are operationalised in the empirical model.

and social influence processes. This macro–micro integration allows the model to capture both structural influences and individual sense-making processes underlying generative AI adoption. Although performance expectancy and effort expectancy are central constructs in the original UTAUT model, in the present integrated framework (shown in Table 1 and Fig. 1), they are conceptually reflected through perceived usefulness and perceived ease of use, consistent with prior integration approaches between UTAUT and TAM2 (Al-kfairy, 2024; Jin et al., 2025). In addition, job relevance, output quality, and result demonstrability correspond to performance expectancy (Asag et al., 2024, pp. 483–489). Within TAM2 and UTAUT, social influence is not treated as a monolithic construct but is theorised to operate through normative pressure (Subjective Norm) and social identity considerations (Image). Accordingly, these two constructs are conceptualised as complementary dimensions of social influence, capturing both compliance-based and identity-based mechanisms of adoption.

Although *Facilitating Conditions* is a core determinant in the UTAUT model, this study does not measure it directly. Instead, it is a contextual boundary condition reflecting infrastructural and institutional realities within Bangladeshi higher education. In developing-country environments, where formal technological support and policy frameworks are often limited, facilitating conditions manifest indirectly through students' perceptions of usefulness, ease of use, and demonstrability, as well as through peer-driven learning networks that substitute for institutional scaffolding (Anik & Rahman, 2025; Saidu & Mamun, 2025). Accordingly, while the empirical model focuses on TAM2's measurable constructs, the influence of facilitating conditions is conceptually incorporated as an underlying environmental moderator that shapes both cognitive evaluations and social influence processes in generative AI adoption. This alignment advances theory by offering a multi-layered socio-cognitive framework that integrates evaluative beliefs and social-normative pressures, providing a context-sensitive model particularly suited for understanding generative AI adoption in resource-constrained, collectivist settings such as Bangladesh.

It is important to note that structural barriers prevalent in developing-country contexts, such as limited device access, inadequate faculty training in AI tools, and the absence of formal institutional AI policies, are not excluded from this study's conceptual scope but are rather reframed as constitutive elements of the facilitating conditions environment. These barriers are treated as macro-level contextual constraints that shape the upper boundary within which students' cognitive and social evaluations of GenAI tools operate. Their influence is theorised to manifest through depressed perceptions of ease of use, reduced result demonstrability, and weakened subjective norms, rather than as independently modelled predictors. This treatment is consistent with

theoretical guidance on contextual boundary conditions in technology acceptance research (Venkatesh et al., 2003) and reflects the empirical realities of conducting model-based research in environments where institutional support remains inconsistent and difficult to operationalise uniformly.

Accordingly, Fig. 1 presents the proposed integrated UTAUT–TAM2 model, which combines cognitive instrumental factors and social influence mechanisms to explain students' adoption of generative AI.

It is important to note that the proposed UTAUT–TAM2 integration is not inherently restricted to engineering education. Rather, it represents a general socio-cognitive framework for understanding generative AI adoption in academic contexts. In this study, engineering education serves as a theoretically meaningful instantiation due to its strong emphasis on task performance, applied problem-solving, and demonstrable outcomes. As such, while the structural relationships of the model are expected to hold across disciplines, the salience and interpretation of specific constructs may vary depending on disciplinary practices and pedagogical norms.

4. Methodology

4.1. Research context and sampling technique

The study was conducted at the Islamic University of Technology (IUT), an international university in Bangladesh recognised for its emphasis on engineering and technical education within a culturally diverse learning environment. Hosting students from South Asia, the Middle East, and Africa, IUT offers a distinctive context for examining how social and cognitive factors interact in the adoption of generative AI. Engineering students at IUT regularly engage in computational modelling, simulation, and design projects that demand creative and analytical reasoning that align closely with the pedagogical value of AI-based tools such as ChatGPT. This environment, therefore, provides an ideal context to explore how variations in cultural background, technical competence, and academic experience shape AI adoption behaviours in a low resource yet globally connected institution.

In addition, the international student cohort at IUT represents a structurally unique population shaped by the university's mandate under the Organisation of Islamic Cooperation (OIC). Students are recruited from over 50 member states spanning sub-Saharan Africa (e.g., Nigeria, Comoros, Uganda, Somalia, Sierra Leone etc.), South Asia (e.g., Pakistan, Afghanistan, Maldives and so on), and the Middle East (e.g., Palestine and Yemen), typically through government nomination schemes and supported by full or partial scholarships covering tuition, accommodation, and living expenses. The medium of instruction at IUT

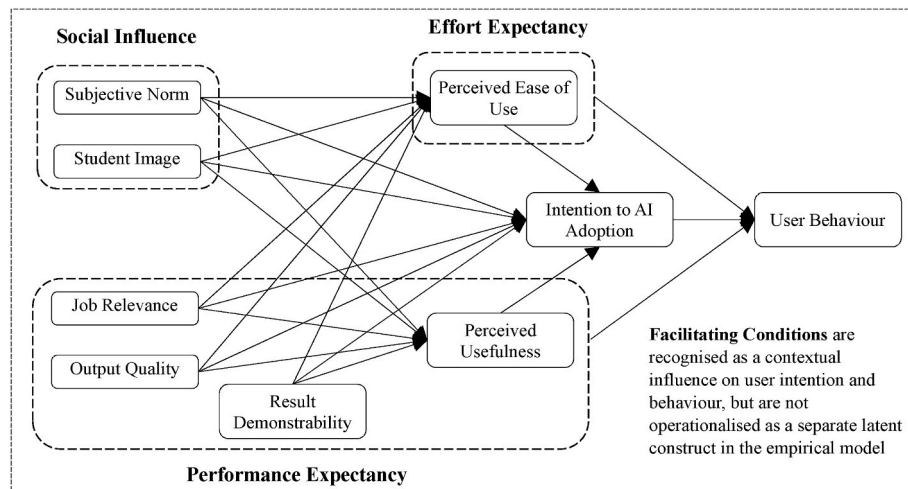


Fig. 1. The socio-cognitive framework for GenAI adoption based on UTAUT–TAM2 integration.

is English; however, students arrive with diverse linguistic backgrounds, prior educational systems, and varying levels of exposure to English-medium technical instruction. This diversity introduces heterogeneity in academic adaptation, communication practices, and digital tool usage. Consequently, international students may differ from domestic students not only in cultural orientation but also in their familiarity with AI-driven academic tools, access to prior training, and confidence in interacting with English-dominant generative AI systems. Incorporating this contextual understanding is essential for interpreting group-level differences in AI adoption behaviour observed in the study.

A two-stage sampling strategy was adopted to ensure both relevance and inclusiveness. In the first stage, purposive sampling targeted undergraduate and graduate engineering students likely to have prior exposure to generative AI applications. In the second stage, convenience sampling was employed to reach a broader pool through institutional mailing lists, departmental groups, and student communication platforms such as WhatsApp and social media. Although convenience sampling may limit generalizability, proactive recruitment across departments and academic levels enhanced representativeness and diversity. Table 2 shows that a total of 378 valid responses were obtained, comprising 155 female (41.0%) and 223 male (59.0%) students. The sample included 205 domestic (54.2%) and 173 international (45.8%) participants, reflecting IUT's multicultural composition and reinforcing the sample's suitability for investigating cross-cultural dynamics in AI adoption.

4.2. Instrument development

The corresponding measurement items for the survey were primarily adapted from Venkatesh and Davis (2000) and subsequently contextualised using validated scales from studies on AI-based educational technologies (Almogren et al., 2024; Oc et al., 2025; Sallam et al., 2024; Shahzad et al., 2024). Particular attention was given to ensuring alignment with educational settings in developing regions and to reflecting the social image, peer influence, and attitudinal dimensions relevant to generative AI adoption. All items in the survey were measured on a five-point Likert scale on a continuum from strongly disagree to strongly agree.

Items measuring Subjective Norms, Job Relevance, Output Quality, and Result Demonstrability were adapted from TAM2 and UTAUT studies to capture students' perceptions of social and task-oriented factors influencing adoption (Sallam et al., 2024; Shahzad et al., 2024; Venkatesh & Davis, 2000; Venkatesh et al., 2003). Consistent with prior UTAUT-TAM2 research, Subjective Norm and Image were selected to capture the normative and reputational dimensions of social influence, rather than broader psychological or ethical constructs. Perceived Usefulness and Perceived Ease of Use were grounded in the original TAM (Davis, 1989; Sallam et al., 2024; Shahzad et al., 2024), whereas Attitude Toward Use and Usage Behaviour were adapted from studies on technology adoption in academic contexts (Davis, 1989; Oc et al., 2025; Sallam et al., 2024; Shahzad et al., 2024; Venkatesh & Davis, 2003). Student Image items were refined to represent reputational and motivational aspects of AI adoption, consistent with the 'image' construct in TAM2.

The initial version of the questionnaire was reviewed by three experts in educational technology in learning to ensure content validity, clarity, and contextual relevance. Minor modifications were made to

align terminologies with generative AI applications, e.g., ChatGPT, AI-powered tools, and generative AI chatbots. A pilot test with 40 engineering students confirmed the clarity of items and internal consistency of the constructs, with all Cronbach's alpha values exceeding the 0.70 threshold. This rigorous multi-stage adaptation and validation process ensured that the final instrument reliably captured both the social and cognitive determinants of generative AI adoption among engineering students in Bangladesh.

5. Data analysis techniques

To examine the relationships among the constructs and validate the hypothesised framework, a multi-stage analytical strategy was employed using Partial Least Squares Structural Equation Modelling (PLS-SEM) with SmartPLS 4 software (Ringle et al., 2024). Following the two-step approach proposed by Anderson and Gerbing (1988), the analysis first assessed the measurement model to confirm the reliability and validity of the observed indicators and then evaluated the structural model to test the hypothesised relationships among the latent constructs. This approach aligns with established best practices in recent technology acceptance research (Almogren et al., 2024; Foroughi et al., 2024; Zhao et al., 2024), ensuring methodological rigour and robustness in analysing both direct and mediated effects within the context of generative AI adoption in higher education.

To examine potential group differences in structural relationships such as gender (male vs. female) and student status (local vs. international), multi-group analysis (MGA) was conducted using the permutation approach implemented in SmartPLS. Prior to MGA, measurement invariance across groups was assessed using the Measurement Invariance of Composite Models (MICOM) procedure, including configural invariance, compositional invariance, and equality of composite mean values and variances. Establishing the partial or full measurement invariance allowed for a valid comparison of path coefficients across groups.

6. Data analysis and results

6.1. Descriptive statistics of the theoretical constructs

Table 3 presents the descriptive statistics for all latent constructs in the study. The mean values indicate positive perceptions toward generative AI across dimensions. Standard deviations were moderate, suggesting reasonable variability in responses without excessive dispersion. Skewness and kurtosis values for all constructs fell within the acceptable range (± 1), implying approximate normality at the univariate level. Overall, the descriptive patterns align with expectations from prior technology adoption literature, supporting the reliability of the dataset for subsequent measurement and structural model analysis.

6.2. Age-construct associations by gender

To examine whether age relates differently to key socio-cognitive constructs across gender, we estimated Pearson correlations (r) separately for female and male students, with 95% confidence intervals (CIs). Fig. 2 presents estimation plots for the associations between age and (a) Subjective Norm (SN), (b) Student Image (SI), (c) Intention to Adopt AI (IAA), and (d) User Behaviour (UB). The figure also reports the estimated gender difference in correlations ($\Delta r = r_{\text{male}} - r_{\text{female}}$) with corresponding CIs.

Across all four constructs, age demonstrated consistently negative associations among female engineering students, ranging from small to moderate magnitude ($|r| \approx .25\text{--}0.38$). In contrast, associations among male students were uniformly weak ($|r| \leq .12$) and in several cases statistically indistinguishable from zero. The gender differences in correlations ($\Delta r \approx 0.18\text{--}0.26$) were consistently positive, indicating that the negative age effects observed among females were substantially

Table 2
Demographic information of the participants.

Demographic Info	Category	Counts	% of Total	Cumulative %
Gender	Female	155	41.01%	41.01%
	Male	223	58.99%	100.0%
Student Status	Domestic	205	54.23%	54.23%
	International	173	45.77%	100.0%

Table 3
Descriptive analysis.

Constructs	Mean	SD	Min	Max	Skewness		Kurtosis	
					Skewness	SE	Kurtosis	SE
SN	14.8	2.72	6	20	−0.495	0.125	0.459	0.250
PU	16.4	2.77	6	20	−0.826	0.125	0.660	0.250
PEOU	12.1	2.31	3	15	−0.804	0.125	0.770	0.250
SI	18.4	3.97	8	25	−0.280	0.125	−0.638	0.250
RD	11.8	2.11	5	15	−0.686	0.125	0.353	0.250
OQ	11.0	2.35	4	15	−0.306	0.125	−0.125	0.250
JR	11.9	2.20	3	15	−0.654	0.125	0.352	0.250
IAA	16.4	3.01	6	20	−1.013	0.125	0.806	0.250
BU	16.1	3.05	5	20	−0.759	0.125	0.448	0.250

Note: The standard errors of skewness and kurtosis are identical across constructs because these values are determined solely by sample size.

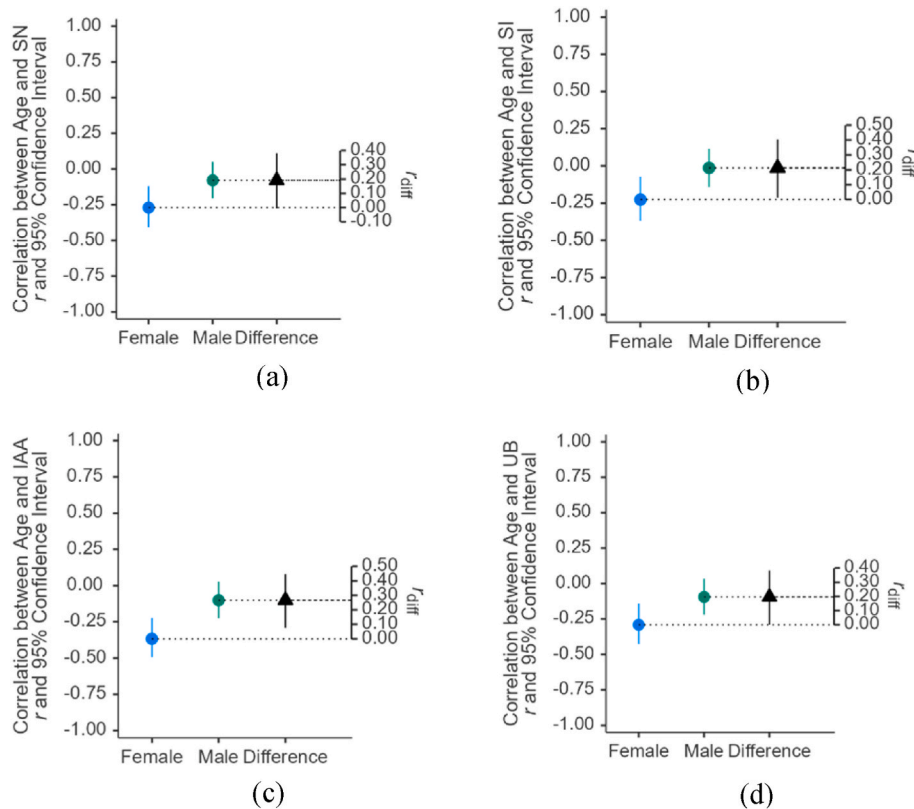


Fig. 2. Correlation between age and (a) Subjective Norm, (b) Student Image, (c) Intention to Adopt AI, and (d) User Behaviour.

attenuated among males.

Substantively, age is more strongly and consistently associated with reduced social pressure, image concern, adoption intention, and AI-related behaviour among female students than among males, for whom these associations are weak or negligible. Increasing age among female engineering students corresponds to lower normative influence systematically, diminished image salience, weaker adoption intention, and reduced behavioural engagement, indicating a clear gendered age gradient in readiness for Generative AI integration. Within the proposed Unified Socio-Cognitive Model, age thus operates as a gender-contingent contextual variable, moderating both social (SN, SI) and cognitive-behavioural (IAA, UB) pathways to Generative AI adoption. Collectively, the results underscore the importance of disaggregated analyses in engineering education research and suggest that interventions aimed at sustaining AI engagement among older cohorts may require particular attention to female students.

6.3. Assessing measurement model

6.3.1. Reliability and convergent validity

The reliability and convergent validity of the measurement model were assessed to ensure that each construct accurately reflects its theoretical domain. Following the recommendations of Hair et al. (2017), four key indicators were examined: item loadings, Cronbach's alpha (α), composite reliability (CR), and average variance extracted (AVE). These complementary indices collectively establish the constructs' internal consistency and measurement quality estimated through PLS-SEM.

As presented in Table 4, all constructs exhibited Cronbach's alpha values above the conventional 0.70 threshold, confirming satisfactory internal consistency. The only exception was Output Quality ($\alpha = 0.671$), which fell marginally below the benchmark due to relatively lower inter-item correlations influenced by OQ1 (accuracy and reliability of AI-generated responses). Despite this, the item was retained owing to its theoretical importance in representing a critical dimension

Table 4
Reliability and Convergent validity.

Construct reliability and validity	Cronbach's alpha	Composite reliability	AVE
Intention to AI Adoption	0.851	0.851	0.691
Job Relevance	0.729	0.731	0.649
Output Quality	0.671	0.746	0.585
Perceived Ease of Use	0.794	0.797	0.707
Perceived Usefulness	0.754	0.757	0.575
Result Demonstrability	0.731	0.750	0.650
Students' Image	0.824	0.835	0.586
Subjective Norms	0.710	0.723	0.632
User Behaviour	0.804	0.820	0.718

of perceived output quality. Importantly, the construct demonstrated acceptable CR (0.746) and AVE (0.585) values, confirming overall construct reliability and convergent validity.

Composite reliability values for all constructs exceeded the minimum requirement of 0.70, with most reflecting good to excellent reliability. Similarly, all AVE scores surpassed the 0.50 threshold, indicating that the latent variables explained more than half of the variance in their indicators (Hair et al., 2017). Item loadings were generally strong (>0.70), and one weak item under Subjective Norms was removed during model refinement to improve measurement precision. The remaining items demonstrated high factor loadings and theoretical coherence, ensuring that each construct retained both conceptual integrity and statistical robustness. Overall, the results confirm that the measurement model exhibits satisfactory reliability and convergent validity, providing a stable foundation for subsequent assessment of discriminant validity and structural relationships among constructs.

6.3.2. Discriminant validity

Discriminant validity ensures that constructs within a measurement model are empirically distinct and not excessively correlated, thereby confirming that each represents a unique conceptual domain (Fornell & Larcker, 1981; Henseler et al., 2015). In this study, discriminant validity was evaluated using three complementary methods: the Fornell–Larcker criterion, cross-loadings, and the Heterotrait–Monotrait (HTMT) ratio. First, the Fornell–Larcker criterion confirmed that the square root of the average variance extracted (AVE) for each construct exceeded its highest correlation with any other construct, as displayed along the diagonals in Table 5. This pattern indicates that each construct shared more variance with its own indicators than with other constructs, thus supporting discriminant validity across the measurement model.

Second, the cross-loadings of individual items were examined to ensure that each indicator loaded more strongly on its intended construct than on any other latent variable. We found all items demonstrated clear loading differentiation, confirming that the survey indicators accurately reflected their theoretical constructs and avoided problematic cross-dimensional contamination (Hair et al., 2017). Third, the Heterotrait–Monotrait (HTMT) ratio was used to provide a more stringent assessment of construct distinctiveness (Henseler et al., 2015). As shown in Table 6, all HTMT values were below the conservative 0.90 threshold, except for two construct pairs, Intention to AI Adoption (IAA)

with User Behaviour (UB), and IAA with Job Relevance (JR), which slightly exceeded the threshold (0.915 and 0.919, respectively). However, bias-corrected confidence interval analyses revealed that none of these intervals contained the critical value of 1.00, indicating that the constructs remain empirically distinct despite marginally elevated correlations. This finding is theoretically consistent, as both IAA and UB reflect sequential stages in behavioural intention and technology use.

6.4. Assessing structural model

Before testing the hypothesised relationships, the structural model was evaluated for potential multicollinearity and predictive explanatory power (R^2). Fig. 3 shows the structural relationships and path coefficients of the proposed model. As presented in Table 7, the VIF values in the current model ranged between 1.426 and 2.587, confirming that the predictors were sufficiently independent and that the model maintained statistical stability. According to Hair et al. (2017), VIF values above 5.0 may indicate potential multicollinearity, while values exceeding 10 represent a serious concern (Kline, 2016).

The model's explanatory power was evaluated using the coefficient of determination (R^2), indicating the variance in each endogenous construct explained by its predictors. Table 7 reports collinearity diagnostics (VIF) and explained variance (R^2) for endogenous constructs. As shown in Fig. 3 and Table 7, the model explains 64.0% of the variance in User Behaviour (UB), 46.1% in Perceived Ease of Use (PEOU), 65.1% in Intention to Adopt AI (IAA), and 53.9% in Perceived Usefulness (PU). These variances are mainly driven by combinations of Intention to Adopt AI, Job Relevance, Output Quality, Result Demonstrability, Student's Image, Subjective Norms, and Perceived Usefulness/Ease of Use. According to Hair et al. (2017), R^2 values of 0.19, 0.33, and 0.67 denote weak, moderate, and substantial explanatory power, respectively. Thus, the results indicate that the proposed model demonstrates moderate to strong predictive capability, confirming its structural and theoretical adequacy in explaining generative AI adoption in higher education.

6.5. Testing hypotheses

To evaluate the hypothesised structural relationships, path coefficients (β), t-values, p-values, and effect sizes (f^2) were estimated using bootstrapping with 5000 resamples (Hair et al., 2017). Practical significance was assessed using Cohen, Manion, Morrison, and 7th's (2013) f^2 thresholds (0.02 = small, 0.10 = medium, 0.35 = large). The results are organised into three sub-sections: direct effects, indirect and mediation effects, and total effects.

6.5.1. Direct effects

As shown in Table 8, 16 of 20 hypothesised paths were significant, offering strong support for the structural model. Subjective Norm (SN), Job Relevance (JR), and Result Demonstrability (RD) strongly influenced Perceived Usefulness (PU) and Perceived Ease of Use (PEOU), which in turn predicted Intention to Adopt AI (IAA) and Usage Behaviour (UB). The most pronounced effects were IAA $\rightarrow$ UB ($\beta = 0.543$, $p < 0.001$), PEOU $\rightarrow$ UB ($\beta = 0.245$, $p < 0.001$), and PU $\rightarrow$ IAA ($\beta = 0.270$, p

Table 5
Latent variable correlations with fornell-larcker criterion.

	IAA	JR	OQ	PEOU	PU	RD	SI	SN	UB
IAA	0.831								
JR	0.725	0.805							
OQ	0.574	0.648	0.765						
PEOU	0.619	0.609	0.544	0.841					
PU	0.697	0.656	0.578	0.607	0.758				
RD	0.641	0.634	0.636	0.551	0.583	0.806			
SI	0.521	0.611	0.617	0.456	0.570	0.562	0.765		
SN	0.522	0.485	0.402	0.494	0.510	0.495	0.310	0.795	
UB	0.767	0.675	0.566	0.644	0.632	0.659	0.536	0.483	0.847

Table 6
Results of Heterotrait-Monotrait (HTMT) Ratio for discriminant validity.

	IAA	JR	OQ	PEOU	PU	RD	SI	SN	UB
IAA	-								
JR	0.919	-							
OQ	0.670	0.885	-						
PEOU	0.747	0.794	0.710	-					
PU	0.872	0.882	0.742	0.776	-				
RD	0.799	0.857	0.823	0.722	0.777	-			
SI	0.620	0.777	0.803	0.542	0.701	0.735	-		
SN	0.664	0.674	0.509	0.659	0.687	0.684	0.404	-	
UB	0.915	0.879	0.740	0.797	0.790	0.865	0.648	0.639	-

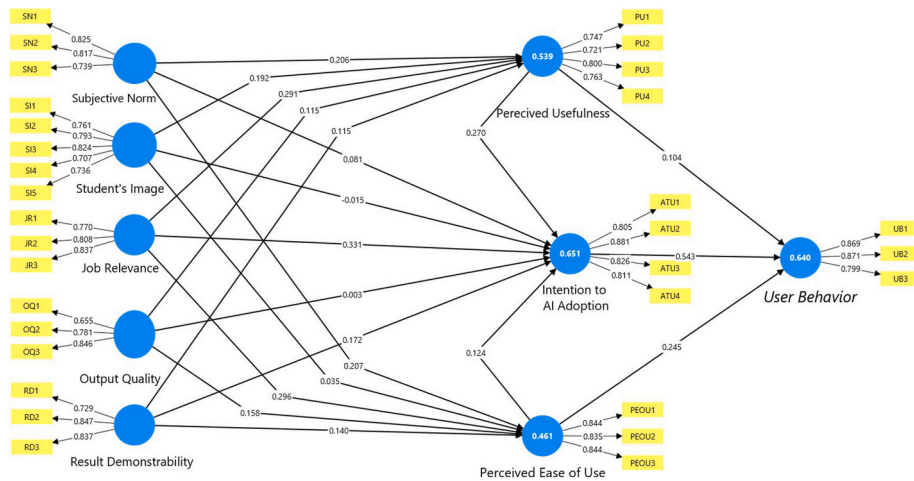


Fig. 3. Results of the structural model.

Table 7
Results of the structural model.

Endogenous construct	Relationship	VIF	R ²
Perceived Ease of Use (PEOU)	JR → PEOU	2.305	0.461
	OQ → PEOU	2.210	
	RD → PEOU	2.159	
	SI → PEOU	1.909	
Perceived Usefulness (PU)	JR → PU	2.305	0.539
	OQ → PU	2.210	
Intention to Adopt AI (IAA)	JR → IAA	2.587	0.651
	OQ → IAA	2.271	
	PEOU → IAA	1.961	
	PU → IAA	2.291	
User Behaviour (UB)	RD → IAA	2.212	0.640
	SI → IAA	1.990	
	SN → IAA	1.566	
	IAA → UB	2.207	
	PEOU → UB	1.796	
	PU → UB	2.157	

< 0.001).

These findings reaffirm the TAM premise that behavioural intention is the primary driver of technology use. JR and Output Quality further strengthened PU and PEOU, while Student Image (SI) showed no significant influence, indicating limited relevance of social prestige in academic contexts. Overall, the results validate the model's predictive robustness, highlighting the central roles of cognitive appraisals and contextual enablers in generative AI adoption.

6.5.2. Indirect and mediation effects

To examine whether exogenous effects on behavioural outcomes operated through Perceived Usefulness (PU), Perceived Ease of Use (PEOU), and Intention to Adopt AI (IAA), mediation analysis was

Table 8
Direct effects: Hypothesis testing results.

Relationship	Std. (β)	Std. error	t value	P values	Decision	f ²
SN → PU	0.206	0.046	4.490***	0.0	Supported	0.064
SN → IAA	0.081	0.041	1.992*	0.046	Supported	0.012
SN → PEOU	0.207	0.055	3.798***	0.0	Supported	0.056
SI → PU	0.192	0.045	4.264***	0.0	Supported	0.042
SI → IAA	-0.015	0.039	0.388	0.698	Not Supported	0.000
SI → PEOU	0.035	0.058	0.605	0.545	Not Supported	0.000
JR → PU	0.291	0.056	5.195***	0.0	Supported	0.080
JR → IAA	0.331	0.062	5.299***	0.0	Supported	0.121
JR → PEOU	0.296	0.071	4.169***	0.0	Supported	0.070
OQ → PU	0.115	0.052	2.214*	0.027	Supported	0.013
OQ → IAA	0.003	0.049	0.057	0.954	Not Supported	0.000
OQ → PEOU	0.158	0.07	2.256*	0.024	Supported	0.021
RD → PU	0.115	0.061	1.884	0.06	Not Supported	0.013
RD → IAA	0.172	0.06	2.868**	0.004	Supported	0.039
RD → PEOU	0.14	0.056	2.514*	0.012	Supported	0.017
PU → IAA	0.27	0.05	5.415***	0.0	Supported	0.092
PEOU → IAA	0.124	0.05	2.459*	0.014	Supported	0.022
PU → UB	0.104	0.042	2.467*	0.014	Supported	0.014
PEOU → UB	0.245	0.044	5.543***	0.0	Supported	0.093
IAA → UB	0.543	0.052	10.521***	0.0	Supported	0.371

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Standardised path coefficients reported. f^2 = Cohen's effect size (0.02 = small, 0.10 = medium, 0.35 = large). Paths marked "Not Supported" indicate no significant direct effect; their influence may operate indirectly through mediating constructs (see Table 9). These direct effects represent the immediate influence of exogenous and mediating constructs on endogenous variables, independent of indirect pathways.

Table 9

Indirect effects: Mediation analysis results.

Relationship	Std. (β)	Std. error	t value	P value	Decision
SN $\rightarrow$ PU $\rightarrow$ IAA	0.056	0.016	3.487***	0.0	Partial mediation
SN $\rightarrow$ PU $\rightarrow$ IAA $\rightarrow$ UB	0.03	0.009	3.309**	0.001	Partial mediation
SN $\rightarrow$ PU $\rightarrow$ UB	0.021	0.01	2.117*	0.034	Partial mediation
SN $\rightarrow$ IAA $\rightarrow$ UB	0.044	0.022	1.989*	0.047	Partial mediation
SN $\rightarrow$ PEOU $\rightarrow$ IAA	0.026	0.012	2.137*	0.033	Partial mediation
SN $\rightarrow$ PEOU $\rightarrow$ IAA $\rightarrow$ UB	0.014	0.007	2.086*	0.037	Partial mediation
SN $\rightarrow$ PEOU $\rightarrow$ UB	0.051	0.018	2.881**	0.004	Partial mediation
SI $\rightarrow$ PU $\rightarrow$ IAA	0.052	0.017	3.131**	0.002	Full mediation
SI $\rightarrow$ PU $\rightarrow$ IAA $\rightarrow$ UB	0.028	0.009	3.043**	0.002	Partial mediation
SI $\rightarrow$ PU $\rightarrow$ UB	0.02	0.009	2.158*	0.031	Partial mediation
SI $\rightarrow$ IAA $\rightarrow$ UB	-0.008	0.021	0.387	0.699	No mediation
SI $\rightarrow$ PEOU $\rightarrow$ IAA	0.004	0.008	0.548	0.584	No mediation
SI $\rightarrow$ PEOU $\rightarrow$ UB	0.009	0.015	0.588	0.557	No mediation
JR $\rightarrow$ PU $\rightarrow$ IAA	0.079	0.021	3.767***	0.0	Partial mediation
JR $\rightarrow$ PU $\rightarrow$ IAA $\rightarrow$ UB	0.043	0.012	3.579***	0.0	Partial mediation
JR $\rightarrow$ PU $\rightarrow$ UB	0.03	0.013	2.338*	0.019	Partial mediation
JR $\rightarrow$ IAA $\rightarrow$ UB	0.18	0.042	4.275***	0.0	Partial mediation
JR $\rightarrow$ PEOU $\rightarrow$ IAA	0.037	0.016	2.233*	0.026	Partial mediation
JR $\rightarrow$ PEOU $\rightarrow$ IAA $\rightarrow$ UB	0.02	0.009	2.200*	0.028	Partial mediation
JR $\rightarrow$ PEOU $\rightarrow$ UB	0.072	0.021	3.472***	0.001	Partial mediation
OQ $\rightarrow$ PU $\rightarrow$ IAA	0.031	0.016	1.954	0.051	No mediation
OQ $\rightarrow$ PU $\rightarrow$ IAA $\rightarrow$ UB	0.017	0.009	1.947	0.052	No mediation
OQ $\rightarrow$ PU $\rightarrow$ UB	0.012	0.008	1.449	0.147	No mediation
OQ $\rightarrow$ IAA $\rightarrow$ UB	0.002	0.027	0.057	0.955	No mediation
OQ $\rightarrow$ PEOU $\rightarrow$ IAA	0.02	0.014	1.432	0.152	No mediation
OQ $\rightarrow$ PEOU $\rightarrow$ UB	0.039	0.019	2.077*	0.038	Partial mediation
RD $\rightarrow$ PU $\rightarrow$ IAA	0.031	0.017	1.810	0.07	No mediation
RD $\rightarrow$ PU $\rightarrow$ UB	0.012	0.009	1.314	0.189	No mediation
RD $\rightarrow$ IAA $\rightarrow$ UB	0.094	0.033	2.827**	0.005	Partial mediation
RD $\rightarrow$ PEOU $\rightarrow$ IAA	0.017	0.01	1.728	0.084	No mediation
RD $\rightarrow$ PEOU $\rightarrow$ UB	0.034	0.015	2.254*	0.024	Partial mediation
PU $\rightarrow$ IAA $\rightarrow$ UB	0.147	0.029	5.107***	0.0	Partial mediation
PEOU $\rightarrow$ IAA $\rightarrow$ UB	0.067	0.028	2.403*	0.016	Partial mediation

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. Full mediation = significant indirect path with non-significant direct path. Partial mediation = both direct and indirect paths significant. Bootstrapping with 5000 resamples used for all estimates.

conducted using bootstrapping with 5000 resamples (Hair et al., 2017). Mediation type was classified as full mediation where the direct path was non-significant but the indirect path was significant, and as partial mediation where both direct and indirect paths were significant.

As shown in Table 9, several significant indirect effects confirmed the mediating roles of cognitive and attitudinal variables. PU and IAA mediated the link between Subjective Norm (SN) and Usage Behaviour (UB), indicating that social influence acts mainly through usefulness perceptions and adoption intention. Job Relevance (JR) also exerted consistent indirect effects via PU, PEOU, and IAA, emphasizing its role in

shaping ease-of-use beliefs and behavioural intentions. In contrast, Student Image (SI) and Output Quality (OQ) showed weak or non-significant mediation. The significant PU $\rightarrow$ IAA $\rightarrow$ UB and PEOU $\rightarrow$ IAA $\rightarrow$ UB pathways align with TAM's hierarchical logic, where cognitive beliefs drive attitudes that, in turn, predict actual use.

The mediation structure of the model reflects a hierarchical socio-cognitive process in which external factors (e.g., Subjective Norm, Job Relevance, Result Demonstrability) influence behavioural outcomes through successive cognitive and attitudinal stages. Specifically, Perceived Usefulness and Perceived Ease of Use function as primary cognitive mediators, while Intention to Adopt AI represents a proximal behavioural mediator linking perceptions to actual usage. Based on the significance of both direct and indirect effects, the results indicate that most relationships exhibit partial mediation, where both direct and indirect paths are significant (e.g., SN $\rightarrow$ IAA $\rightarrow$ UB; JR $\rightarrow$ IAA $\rightarrow$ UB). In contrast, instances such as Student Image (SI), where the direct effect on intention is non-significant but indirect effects via PU are significant, reflect full mediation, suggesting that its influence operates entirely through cognitive appraisal mechanisms. Mediation results show that user beliefs (PU, PEOU) and intentions (IAA) are key pathways linking contextual and social factors (JR, SN, RD) to system use, reinforcing the model's robustness in explaining generative AI adoption in higher education.

6.5.3. Total effects

Table 10 reports the total effects (i.e., the sum of direct and indirect effects) of all predictors on Usage Behaviour (UB), to provide a comprehensive picture of each construct's cumulative influence on the model's primary outcome variable. The results indicate that constructs such as Job Relevance (JR) and Subjective Norm (SN) exert substantial total effects on Usage Behaviour (UB), primarily through mediated pathways involving PU, PEOU, and IAA.

Collectively, these total effects affirm that while intention remains the most proximal determinant of usage, the upstream influence of task-relevance and social-normative factors is substantially underestimated when only direct effects are considered. This reinforces the interpretation that while some variables may exhibit modest direct effects, their overall influence on behaviour is amplified through the socio-cognitive mediation structure of the model.

6.6. Multi-group analysis and measurement invariance

Measurement invariance was assessed using the MICOM procedure in SmartPLS 4, following the guidelines of Henseler et al. (2016). Configural invariance was established across all groups. Compositional invariance was evaluated using 1000 permutations, and partial measurement invariance was achieved in both comparisons, satisfying the prerequisite for multi-group analysis (MGA). Table 11 presents the MICOM results for the gender comparison, indicating the level of

Table 10

Total effects on usage behaviour.

Relationship	Direct Effect (β)	Indirect Effect (β)	Total Effect (β)	Mediation Status
SN $\rightarrow$ UB	—	0.160	0.160	Full mediation
SI $\rightarrow$ UB	—	0.048	0.048	Full mediation
JR $\rightarrow$ UB	—	0.345	0.345	Full mediation
OQ $\rightarrow$ UB	—	0.039	0.039	Full mediation
RD $\rightarrow$ UB	—	0.128	0.128	Full mediation
PU $\rightarrow$ UB	0.104	0.147	0.251	Partial mediation
PEOU $\rightarrow$ UB	0.245	0.067	0.312	Partial mediation
IAA $\rightarrow$ UB	0.543	—	0.543	No mediation

Note: Total Effect = Direct Effect + Indirect Effect; Full mediation: Direct effect is non-significant, but indirect is significant; Partial mediation: Both direct and indirect effects are significant; No mediation: Only direct effect is significant.

Table 11
MICOM results for gender.

Construct	Configural Invariance	Compositional Invariance (Step 2)	Mean Equality	Variance Equality	Invariance Established
IAA	Yes	Yes (p = 0.995)	Yes (p = 0.182)	No (p = 0.006)	Partial
JR	Yes	Yes (p = 0.053)	Yes (p = 0.451)	Yes (p = 0.229)	Full
OQ	Yes	Yes (p = 0.632)	Yes (p = 0.474)	Yes (p = 0.094)	Full
PEOU	Yes	Yes (p = 0.385)	Yes (p = 0.164)	Yes (p = 0.791)	Full
PU	Yes	Yes (p = 0.128)	Yes (p = 0.654)	Yes (p = 0.717)	Full
RD	Yes	Yes (p = 0.270)	Yes (p = 0.076)	No (p = 0.001)	Partial
SI	Yes	No (p = 0.042)	Yes (p = 0.256)	Yes (p = 0.412)	None
SN	Yes	Yes (p = 0.134)	Yes (p = 0.978)	Yes (p = 0.364)	Full
UB	Yes	Yes (p = 0.111)	Yes (p = 0.411)	Yes (p = 0.137)	Full

measurement invariance established prior to structural analysis. The compositional invariance was supported for all constructs except SI (p = 0.042). Equality of means was satisfied, while variance equality was not supported for IAA (p = 0.006) and RD (p = 0.001). Given partial invariance, permutation-based MGA was conducted.

Table 12 reports the permutation-based MGA results for gender, comparing structural path coefficients between male and female students. No structural path differences were significant at $p < 0.05$. The relationships among PEOU, PU, IAA, and UB were statistically equivalent across male and female students. Thus, the proposed socio-cognitive model demonstrates structural invariance across gender.

Overall, the permutation MGA results provide evidence of structural

invariance, indicating that gender does not significantly moderate the relationships specified in the unified socio-cognitive model.

Table 13 summarises the MICOM results for the Domestic versus International student groups, confirming partial measurement invariance. For domestic status, compositional invariance was established for all constructs. Although several constructs differed in means or variances, partial invariance was achieved, permitting structural comparison.

Table 14 presents the permutation-based MGA results for domestic and international students, highlighting significant differences in structural relationships between the groups. Permutation MGA identified two significant differences. The effect of PEOU on IAA was stronger among domestic students than international students ($\Delta\beta = 0.282$, $p = 0.001$). This suggests that while domestic users are motivated by the user-friendliness of AI tools, international users' intentions are likely driven by other factors, such as utility or necessity, making ease of use a non-factor in their adoption process. Similarly, SI $\rightarrow$ IAA differed ($\Delta\beta = -0.207$, $p = 0.012$), with a negative effect for domestic students and a positive effect for international students. This flip indicates that the social identity or status associated with being a student using AI provides a motivational boost for international cohorts but may act as a slight deterrent or irrelevant factor for domestic students.

All other paths, including IAA $\rightarrow$ UB (p = 0.822), were invariant. Hence, while the intention-behaviour mechanism is stable across cohorts, ease of use and social influence operate differently depending on domestic status. This indicates that once the intention to adopt AI is formed, the translation of that intention into actual behaviour is consistent regardless of the user's domestic or international status. All other structural pathways (e.g., JR, PU) showed no statistically significant differences between groups ($p > 0.05$).

7. Discussions on the findings

This study explored students' adoption of generative AI technologies in higher education through an integrated TAM2-UTAUT framework, revealing latent relationships among cognitive, social, and contextual factors. The analysis suggests that adoption is not simply a function of perceived usefulness or ease of use but rather is shaped by three inter-related themes: (i) the functional and contextual embeddedness of AI adoption, (ii) the social influence of perceived value, and (iii) digital equity and inequities in AI adoption in multicultural learning environments.

7.1. Functional and contextual embeddedness of AI adoption

Our finding that job relevance (JR) and result demonstrability (RD) strongly predict perceived usefulness (PU) and intention (IAA) underscores that generative AI adoption in engineering education is primarily task-driven rather than technology-driven. This result aligns with TAM/TAM2 logic (Davis, 1989; Venkatesh & Davis, 2000) but adds empirical precision for GenAI, i.e., students adopt when AI demonstrably supports discipline-specific tasks (coding, simulation, debugging) rather than for generic productivity gains. Recent empirical work

Table 12
Permutation-based multi-group analysis results for gender.

	Male (β)	Female (β)	$\Delta\beta$ (M-F)	Permutation p- value	Decision
IAA $\rightarrow$ UB	0.517	0.559	-0.043	0.663	Not Significant
JR $\rightarrow$ IAA	0.249	0.313	-0.065	0.639	Not Significant
JR $\rightarrow$ PEOU	0.363	0.190	0.173	0.247	Not Significant
JR $\rightarrow$ PU	0.271	0.261	0.010	0.929	Not Significant
OQ $\rightarrow$ IAA	-0.013	0.083	-0.096	0.360	Not Significant
OQ $\rightarrow$ PEOU	-0.024	0.273	-0.297	0.064	Not Significant
OQ $\rightarrow$ PU	0.009	0.148	-0.140	0.211	Not Significant
PEOU $\rightarrow$ IAA	0.097	0.086	0.011	0.923	Not Significant
PEOU $\rightarrow$ UB	0.242	0.162	0.080	0.342	Not Significant
PU $\rightarrow$ IAA	0.282	0.273	0.009	0.927	Not Significant
PU $\rightarrow$ UB	0.144	0.120	0.024	0.783	Not Significant
RD $\rightarrow$ IAA	0.166	0.196	-0.030	0.827	Not Significant
RD $\rightarrow$ PEOU	0.148	0.126	0.022	0.861	Not Significant
RD $\rightarrow$ PU	0.110	0.031	0.079	0.575	Not Significant
SI $\rightarrow$ IAA	0.021	-0.050	0.070	0.422	Not Significant
SI $\rightarrow$ PEOU	0.098	-0.074	0.172	0.168	Not Significant
SI $\rightarrow$ PU	0.244	0.213	0.031	0.766	Not Significant
SN $\rightarrow$ IAA	0.177	0.014	0.162	0.058	Not Significant
SN $\rightarrow$ PEOU	0.139	0.212	-0.073	0.572	Not Significant
SN $\rightarrow$ PU	0.293	0.108	0.185	0.074	Not Significant

Note. $\Delta\beta$ = difference in standardised path coefficients (Male - Female). Not significant at $p < 0.05$.

Table 13

MICOM results for students' status.

Construct	Configural Invariance	Compositional Invariance	Mean Equality	Variance Equality	Invariance Established
IAA	Yes	Yes ($p = 0.843$)	No ($p = 0.002$)	Yes ($p = 0.644$)	Partial
JR	Yes	Yes ($p = 0.129$)	No ($p = 0.012$)	Yes ($p = 0.756$)	Partial
OQ	Yes	Yes ($p = 0.255$)	Yes ($p = 0.096$)	Yes ($p = 0.491$)	Full
PEOU	Yes	Yes ($p = 0.641$)	No ($p = 0.001$)	Yes ($p = 0.249$)	Partial
PU	Yes	Yes ($p = 0.964$)	No ($p = 0.002$)	Yes ($p = 0.131$)	Partial
RD	Yes	Yes ($p = 0.176$)	Yes ($p = 0.128$)	Yes ($p = 0.669$)	Full
SI	Yes	Yes ($p = 0.667$)	Yes ($p = 0.595$)	Yes ($p = 0.172$)	Full
SN	Yes	Yes ($p = 0.092$)	Yes ($p = 0.439$)	No ($p = 0.019$)	Partial
UB	Yes	Yes ($p = 0.654$)	No ($p = 0.013$)	Yes ($p = 0.293$)	Partial

Table 14

Permutation-based multi-group analysis results for students' status.

	Domestic (β)	International (β)	$\Delta\beta$ (D-I)	Permutation p-value	Decision
IAA $\rightarrow$ UB	0.529	0.552	-0.022	0.822	Not Significant
JR $\rightarrow$ IAA	0.133	0.338	-0.205	0.116	Not Significant
JR $\rightarrow$ PEOU	0.291	0.214	0.077	0.602	Not Significant
JR $\rightarrow$ PU	0.221	0.240	-0.019	0.844	Not Significant
OQ $\rightarrow$ IAA	0.020	0.055	-0.035	0.760	Not Significant
OQ $\rightarrow$ PEOU	0.151	0.175	-0.024	0.872	Not Significant
OQ $\rightarrow$ PU	0.108	0.122	-0.014	0.899	Not Significant
PEOU $\rightarrow$ IAA	0.231	-0.051	0.282	0.001	Significant
PEOU $\rightarrow$ UB	0.227	0.176	0.052	0.534	Not Significant
PU $\rightarrow$ IAA	0.355	0.233	0.122	0.152	Not Significant
PU $\rightarrow$ UB	0.112	0.166	-0.054	0.521	Not Significant
RD $\rightarrow$ IAA	0.290	0.114	0.176	0.153	Not Significant
RD $\rightarrow$ PEOU	0.037	0.149	-0.112	0.351	Not Significant
RD $\rightarrow$ PU	0.165	-0.002	0.167	0.236	Not Significant
SI $\rightarrow$ IAA	-0.085	0.121	-0.207	0.012	Significant
SI $\rightarrow$ PEOU	-0.024	0.119	-0.142	0.237	Not Significant
SI $\rightarrow$ PU	0.210	0.245	-0.035	0.722	Not Significant
SN $\rightarrow$ IAA	0.064	0.090	-0.025	0.740	Not Significant
SN $\rightarrow$ PEOU	0.264	0.066	0.198	0.108	Not Significant
SN $\rightarrow$ PU	0.155	0.262	-0.107	0.340	Not Significant

on GenAI in higher education corroborates this task-alignment effect. For example, [Patterson et al. \(2024\)](#) and [Raman \(2024\)](#) document that students are more likely to use ChatGPT and similar systems when they perceive direct curricular and employability benefits.

Critically, output quality (OQ) functioned as a cognitive enabler, influencing perceived usefulness (PU) and perceived ease of use (PEOU) rather than directly affecting intention. This mediational pattern aligns with other applied TAM2 studies, where increased reliability enhances trust and perceived usefulness (PU) but does not directly lead to behavioural intention unless the system is clearly relevant and its outcomes are visible ([Asag & Al Mamun, 2025](#); [Yuan et al., 2021](#)). This suggests a two-stage reasoning that institutions must ensure AI outputs meet a minimum credibility standard (to build PU) and integrate AI tasks into coursework so students can see demonstrable benefits (to convert PU into IAA and UB). The literature on generative AI adoption supports this sequencing. For example, meta-analyses and large survey research indicate that AI tools that clearly improve performance and fit the curriculum are more likely to be adopted and used regularly than those promoted only for their novelty ([Wang & Fan, 2025](#)).

Theoretically, these results refine TAM2 for GenAI by specifying that disciplinary task-fit acts as a moderator of many classic TAM pathways, such as job relevance amplifying the PU $\rightarrow$ IAA link and strengthening the mediating role of PU between OQ and IAA. This moves beyond generic acceptance statements and situates GenAI uptake within practice theory and task-technology fit arguments ([Goodhue & Thompson, 1995](#)), reinforcing calls in the recent literature to operationalise “relevance” in domain-specific terms ([Sergeeva et al., 2025](#)).

The dominance of job relevance (JR) and result demonstrability (RD) as predictors of intention (IAA) confirms that engineering students utilise a cognitive calculus centred on Task-Technology Fit ([Chen et al., 2025](#); [Ismail & Krishnanraw, 2025](#)). Rather than being swayed by the

novelty of GenAI, students engage in a deliberate assessment of performance expectancy, where perceived usefulness (PU) acts as a critical mediating mechanism. This mediation suggests that task-specific attributes only trigger adoption when they are cognitively processed as value-adding for discipline-specific work, such as debugging or simulation. Furthermore, the role of output quality (OQ) as a distal enabler, influencing intent only through the mediators of PU and PEOU, reinforces the TAM2 logic that technical excellence is a prerequisite for trust, but insufficient for adoption without clear curricular alignment ([Alanazi et al., 2025](#); [Hou, 2025](#)). By situating these findings within the broader literature, it becomes evident that GenAI acceptance in technical domains is less about general productivity and more about the precision with which the tool integrates into the cognitive demands of the engineering profession.

7.2. The social shaping of perceived value

Subjective norms (SN) significantly influenced PU and PEOU in our model, showing that social cues remain central in shaping cognitive appraisals of GenAI. This observation supports UTAUT/TAM2 assertions about social influence ([Venkatesh & Davis, 2000](#)) and echoes recent empirical studies showing that instructor and peer endorsement strongly predict students' AI acceptance ([Raman et al., 2024](#); [Shahzad et al., 2024](#); [Strzelecki, 2024](#)). However, the null effect for Student Image (SI) departs from TAM2's proposition that image (status/reputation) can influence perceived usefulness and adoption ([Venkatesh & Davis, 2000](#)) and from empirical studies that find social endorsement matters in prestige-sensitive contexts ([Shahzad et al., 2024](#)). This divergence suggests a contextual boundary condition, indicating that in resource-constrained or highly task-oriented learning environments, instrumental social influences (e.g., peer use, instructor modelling) may

outweigh symbolic social influences (e.g., image or status). Recent empirical work shows that in collectivist and technically oriented learning contexts, conformity to group norms and instructor or peer support are often stronger drivers of AI tool adoption than concerns about personal image or reputation (Almogren et al., 2024; Perera et al., 2025; Shahzad et al., 2024; Sobaih et al., 2024).

Moreover, the mediation analysis suggests that subjective norms exert their influence primarily through perceived usefulness and perceived ease of use. These findings position social influence as a legitimising mechanism that conveys normative acceptance, mitigates perceived uncertainty, and strengthens the formation of cognitive beliefs underlying technology adoption. This understanding is supported by recent qualitative and mixed-methods studies of ChatGPT adoption, showing that peer demonstrations and instructor sanctioning reduce perceived risk and accelerate experimentation (Patterson et al., 2024; Shahzad et al., 2024). Thus, social influence theory in TAM/UTAUT should be elaborated to distinguish legitimisation (endorsed practice leading to cognitive appraisal) from status signalling (student image influencing intention).

Importantly, these results suggest that social influence does not operate in isolation but interacts with students' cognitive evaluations of GenAI. This is evidenced by the significant impact of subjective norms, including peer expectations and institutional attitudes, on adoption intention, a finding supported by recent studies on GenAI integration in higher education (Miranda & Chamorro-Mera, 2025; Rizun et al., 2026). Such normative pressures reflect a broader trend where academic networks and instructor attitudes significantly shape student attitudes toward emerging AI tools. Conversely, student image exhibited a non-significant effect, suggesting that for engineering students, adoption is driven by functional outcomes rather than status signalling. This aligns with recent evidence from technical disciplines where utilitarian benefits, such as task automation and reduced cognitive load, are prioritised over reputational gain (BinJwair, 2025). Together, this socio-cognitive interplay reinforces the integrated UTAUT-TAM2 framework, demonstrating that in structured educational environments, normative pressures shape behaviour primarily by influencing perceived usefulness.

7.3. Differential access: digital equity and inequities in AI adoption

The consistently higher perceptions among domestic students compared to international students (across PU, PEOU, RD, JR, and IAA) point to structural and cultural moderators of adoption that are rarely foregrounded in basic TAM/UTAUT models. This stratification resonates with digital inclusion and access literatures, showing that language familiarity, prior exposure to specific tools, and alignment of curricula with local practice drive differential adoption (Sergeeva et al., 2025).

Importantly, the international student cohort in this study is predominantly drawn from OIC member states, including sub-Saharan African countries such as Nigeria, Tanzania, Uganda, and Sierra Leone, as well as South Asian and Middle Eastern nations such as Pakistan, Sri Lanka, Palestine, and Yemen. Although instruction at IUT is conducted in English, English is not the first language for the majority of these students, and their home educational systems vary considerably in terms of digital infrastructure, AI literacy training, and technology integration practices. These linguistic and socio-educational differences provide a concrete empirical basis for interpreting the observed group-level disparities in adoption perceptions. Specifically, language and cultural barriers may limit international students' ability to fully leverage AI tools, which are designed primarily for English-language academic contexts (Karakas, 2023). Additionally, familiarity with local educational systems may advantage domestic students who better understand how AI tools align with specific course requirements, assessment criteria, and academic expectations (Shahzad et al., 2024). Infrastructural access differences may also contribute to these disparities. Recent GenAI studies echo this equity lens. For instance, Patterson et al.

(2024) and Baharin et al. (2025) note that vocational, TVET contexts and multicultural student populations show variable uptake patterns linked to training availability and contextual relevance, which is consistent with our evidence that domestic students, who may have easier access to local support and language-aligned materials, report stronger adoption indicators.

The MGA results further illuminate how this background differences translate into structural heterogeneity in adoption pathways. The significant cross-group difference in the PEOU $\rightarrow$ IAA path ($\Delta\beta = 0.282$, $p = 0.001$) suggests that for domestic students, the intuitive usability of AI tools is a meaningful driver of adoption intention, likely because their greater English proficiency and familiarity with AI-integrated coursework lower interaction barriers. For international students, however, ease of use appears insufficient to motivate adoption, suggesting that other factors, such as perceived utility under language constraints or necessity-driven use, may govern their intention formation. The opposing directional effect of Student Image on intention (SI $\rightarrow$ IAA: $\Delta\beta = -0.207$, $p = 0.012$) further reflects this divergence: international students appear to derive motivational value from the social identity associated with AI use, possibly as a means of academic legitimisation or peer integration within an unfamiliar institutional environment, whereas domestic students show no such identity-driven impetus.

This observation has two analytical implications. First, it highlights that analysing aggregated TAM/UTAUT data can obscure subgroup differences. Combining all students into a single analysis may inflate the apparent strength of cognitive predictors while masking access barriers faced by minority or international students. Second, it highlights the urgent need to incorporate resource and literacy moderators, e.g., AI literacy, language proficiency, and institutional facilitation, into adoption models. Recent large-scale surveys and meta-analyses of GenAI adoption underscore the same point, i.e., adoption rates and benefits are contingent on training, trust, and contextual readiness (Shahzad et al., 2024; Wang & Fan, 2025).

The total effects analysis further substantiates these equity-related observations. The substantially larger total effect of Perceived Usefulness on Usage Behaviour (total $\beta = 0.251$) compared to its direct effect alone ($\beta = 0.104$), and similarly for Perceived Ease of Use (total $\beta = 0.312$ vs. direct $\beta = 0.245$), highlights that cognitive beliefs exert a broader cumulative influence on actual use than direct effects alone suggest, with IAA serving as the critical mediating bridge. This makes the cross-group difference in the PEOU $\rightarrow$ IAA path ($\Delta\beta = 0.282$, $p = 0.001$) particularly consequential — international students' weaker ease-of-use driven intentions effectively suppress the full cumulative effect of PEOU on UB, contributing to the observed adoption disparities between domestic and international cohorts.

Also, the multi-group analysis reveals a pattern of structural stability and contextual differentiation. First, the absence of significant path differences across gender indicates that the proposed socio-cognitive model operates equivalently for male and female engineering students. The relationships among perceived ease of use, perceived usefulness, intention, and use behaviour are structurally invariant, suggesting that gender does not meaningfully condition generative AI adoption mechanisms within this cohort. The convergence aligns with recent evidence that gender disparities in digital competence and technology engagement are diminishing among digitally native cohorts in higher education (Asag et al., 2024, pp. 483–489). Together, the findings reinforce the view that gender no longer constitutes a salient moderator of AI adoption processes in contemporary academic contexts. In contrast, domestic status introduces selective structural heterogeneity. While the central intention-behaviour linkage remains robust and invariant, two upstream drivers, such as perceived ease of use and social influence, exhibit significant cross-group differences. Specifically, ease of use predicts intention among domestic students but not among international students, whereas social influence demonstrates opposing directional effects. These results indicate that although behavioural enactment processes are stable, intention formation is context-sensitive

and shaped by socio-environmental positioning (Akbarighatar et al., 2026).

Collectively, the findings suggest that generative AI adoption among engineering students is largely gender-neutral but partially contingent upon students' domestic status. Theoretical models of AI adoption in higher education should therefore treat gender as structurally invariant while accounting for contextual moderators linked to institutional and socio-cultural integration.

8. Implications of the study findings

8.1. Data analytics perspectives on pedagogical reform and institutional policy

From an educational data analytics perspective, the findings demonstrate how socio-cognitive constructs derived from the integrated UTAUT–TAM2 framework can be operationalised as analytically meaningful indicators for understanding students' engagement with generative AI tools in higher education. Rather than relying on descriptive adoption rates, the structural and mediating relationships identified in this study provide explanatory insight into *why* students form intentions to adopt generative AI, highlighting the relative salience of cognitive evaluations (e.g., perceived usefulness and ease of use) and socially mediated influences (e.g., subjective norm and image).

Consistent with prior work in learning analytics and educational data mining (Ferguson, 2012; Paolucci et al., 2024), the results suggest that theory-driven analytics can support more informed pedagogical decision-making. For example, the substantial variance explained in intention to adopt AI and user behaviour indicates that constructs such as performance expectancy, job relevance, and social influence can function as early indicators of students' readiness to engage with AI-supported learning activities. These indicators may be leveraged by educators and institutions to design targeted interventions, such as scaffolding activities, peer-supported learning tasks, or transparent guidance on appropriate AI use, rather than adopting uniform or policy-driven approaches.

Importantly, the mediating effects observed in the model illustrate how social and cognitive factors interact dynamically, reinforcing calls within the learning analytics literature to move beyond isolated predictors toward models that capture interdependencies among learner perceptions, social context, and behavioural outcomes (Gašević et al., 2017). From a pedagogical analytics standpoint, this enables institutions to transition from post hoc evaluations of AI adoption toward explanatory and potentially predictive models that can inform curriculum design, assessment practices, and faculty development initiatives.

Furthermore, the identification of group-level differences through multi-group analysis aligns with emerging emphases on equity-aware and context-sensitive analytics in education (Sawair & Ghazzawi, 2025; Senior & Sahlberg, 2025). Understanding how adoption mechanisms vary across demographic groups allows institutions to avoid one-size-fits-all implementations of generative AI and instead adopt data-informed strategies that are responsive to diverse learner profiles. In this sense, the present findings contribute to the growing body of research advocating for analytically grounded, pedagogically meaningful integration of AI technologies in higher education.

The transition toward AI-integrated engineering education requires a strategic shift from viewing technology as a general productivity aid to treating it as a domain-specific cognitive partner. For educators, the primary implication is the need for curricular alignment that prioritises functional utility; because students are driven by task-technology fit, AI should be integrated directly into discipline-specific challenges like debugging, simulation, and technical design, where its relevance is immediate. Furthermore, as instructors act as “normative anchors,” their active modelling of AI use serves to legitimise the technology, reducing the cognitive uncertainty students may feel about its academic validity.

From a policy perspective, institutions must move beyond providing simple access and instead focus on bridging the digital equity gap identified in the multi-group analysis. Since domestic and international students experience different adoption frictions, policies should be tailored to provide context-aware support, such as language-specific prompt engineering training and subsidised access to high-tier models for underrepresented cohorts. Ultimately, institutional frameworks must prioritise faculty upskilling and the development of ethical AI standards to ensure that adoption is not merely a byproduct of individual curiosity but a structured, equitable, and professionally grounded evolution of the engineering curriculum.

8.2. Theoretical contributions, model extension and future directions

This study makes several theoretical contributions by advancing an integrated UTAUT–TAM2 framework specifically for the Generative AI era. While prior models often treat predictors in isolation, this proposed model explicates the specific socio-cognitive mechanisms through which intention is formed in task-intensive environments.

By integrating TAM and UTAUT, this study demonstrates that cognitive and social drivers can be jointly modelled to explain Generative AI (GenAI) adoption with superior precision. This aligns with recent arguments that single-framework models are increasingly insufficient for capturing the multifaceted nature of AI-mediated learning (Al-kfairy, 2024; Rana et al., 2024). Beyond mere model fit, the mediation analysis illuminates the “layered” nature of this process; it reveals how job relevance and social norms influence behaviour indirectly through cognitive evaluations, shifting the focus from a binary “adopt vs. reject” framework toward a socially mediated cognitive process. Specifically, the findings clarify that social influence is not monolithic. By distinguishing between Subjective Norms (which were significant) and Student Image (which was not), the study suggests that in utilitarian fields like engineering, “legitimacy” outweighs “status.” This provides a necessary refinement to TAM2, showing that social constructs operate differently across disciplinary cultures.

Furthermore, by testing this architecture in a multicultural, resource-constrained environment, this research validates the cross-cultural robustness of these constructs within a Global South context. This empirical evidence directly responds to the calls by Ravšelj et al. (2025) to strengthen the external applicability of acceptance theory in emerging economies, where institutional norms differ significantly from Western contexts.

By extending prior integrated acceptance models into the generative AI domain, this study demonstrates how established socio-cognitive frameworks can be adapted to emerging AI-mediated learning practices, thereby advancing both technology acceptance theory and AI-in-education research. While the integrated UTAUT–TAM2 framework provides a robust foundation for understanding adoption intention, the unique nature of Generative AI, characterised by its agency, opacity, and ethical complexity, suggests several critical avenues for model extension.

Integrating Self-Regulated Learning (SRL) and Epistemic Agency: Current models primarily focus on “Performance Expectancy” (the outcome), but GenAI adoption is increasingly tied to how students manage their own learning processes. Future extensions should incorporate Self-Regulated Learning (SRL) constructs to explore whether high adoption intention correlates with deep cognitive engagement or “off-loading” of critical thinking. As noted by Shum (2024), analytics must shift from tracking usage to measuring epistemic agency, ensuring that AI acts as a “cognitive outpost” rather than a substitute for student thought.

Incorporating Moral Norms and Algorithmic Literacy: While this study identified the significance of Subjective Norms, the rapid evolution of academic integrity policies suggests that Moral Norms and Ethical Predispositions should be explicitly modelled. Recent literature, Liu and Lin (2026) suggests that “calibrated trust”—the ability to discern when an AI

is hallucinating—is a critical moderator of long-term adoption. Future models could integrate Algorithmic Literacy as a facilitating condition that determines whether a student's "Ease of Use" leads to productive or counter-productive outcomes.

From Intention to Sustained Human-AI Collaboration: Finally, as GenAI moves from a "novel tool" to a "standard teammate," acceptance models must evolve to capture Human-AI Collaboration (HAIC). This involves moving beyond the binary of "Adopt vs. Reject" to modelling "Synergistic Use." Incorporating constructs from the Theory of Planned Behaviour (TPB), such as perceived behavioural control over AI outputs, would provide a more layered theoretical approach. This would allow researchers to track not just the initial intention to use AI, but the sustained, reflective, and responsible integration of these tools into professional engineering practice (Khosravi et al., 2022).

9. Limitations of the study

Despite its contributions, this study has several limitations that provide a roadmap for future inquiry. First, the cross-sectional design restricts the ability to make definitive causal inferences. While the PLS-SEM results indicate strong associations, reliance on self-reported measures may introduce common method bias—a limitation common in similar technology adoption studies (Strzelecki, 2024). Future research should adopt longitudinal or experimental designs to validate temporal causality and assess how students' reliance on social influence may diminish as they gain technical proficiency.

Second, the study was empirically situated within a single engineering university in Bangladesh. While this provides a contextually rich and deep examination of AI adoption within a task-intensive, resource-constrained, and multicultural academic environment, the institutional, cultural, and infrastructural conditions at IUT represent a specific empirical instantiation of the broader theoretical framework rather than a universal validation of it. Accordingly, the findings should be interpreted as contextually situated, as adoption dynamics may differ across regions, institutional types, and disciplinary contexts. In particular, differences in learning practices and assessment structures in non-engineering disciplines such as the humanities or social sciences may influence how socio-cognitive factors shape adoption in ways not captured by the present model. Cross-institutional, cross-disciplinary, and cross-national replication studies are therefore needed to establish the external validity of the integrated UTAUT-TAM2 framework across diverse higher education contexts.

Third, although the integrated model demonstrates strong explanatory power, it primarily captures socio-cognitive determinants of intention. The current model excluded potentially salient factors such as trust, ethical perception, and anxiety, which recent literature suggests are influential in the generative AI context (Asag & Al Mamun, 2025; Dietrich & Grassini, 2025). As AI increasingly participates in knowledge production rather than mere information delivery, future research should integrate complementary frameworks—such as the Theory of Planned Behaviour (TPB) or Diffusion of Innovations—to account for moral-regulatory reasoning and perceived behavioural control.

Finally, this study operationalised demographic heterogeneity through multi-group analysis (MGA) rather than continuous moderation. While this aligns with established UTAUT practices, future research could utilise interaction-based moderation to capture more fine-grained developmental effects. Furthermore, variables such as digital literacy, year of study, digital literacy, and enrolment type were not explicitly measured. Qualitative or mixed-methods inquiries could unpack these individual differences, exploring how students interpret AI's academic legitimacy and how institutional training shifts attitudes over time.

10. Conclusion

This study advances the understanding of generative AI adoption in higher education by integrating TAM and UTAUT into a unified

framework. By combining socio-cognitive theory with structural modelling, the study contributes to the growing field of educational data analytics by demonstrating how theory-driven analytics can inform pedagogical innovation in AI-enabled learning environments.

The findings reveal that adoption is shaped by the interplay of cognitive evaluations (usefulness, ease of use) and social influences (subjective norms, peer validation), highlighting that students' behavioural intentions toward AI are contextually and socially embedded. The model's strong explanatory power confirms that merging cognitive and social perspectives offers a more comprehensive account of AI use behaviour than either model alone. Furthermore, it extends UTAUT to a developing academic context, demonstrating its robustness outside of traditional Western settings.

Practically, the results suggest that universities can strengthen AI adoption by improving usability, embedding job relevance in curricula, and promoting ethical, inclusive use through targeted institutional support. Despite limitations related to the cross-sectional design and single-site data, the study provides a foundation for future research to explore the evolving perceptions of generative AI through longitudinal or mixed-method approaches. Overall, it positions AI adoption not as a purely technical process but as a socially mediated practice, offering both theoretical refinement and actionable guidance for higher education institutions seeking responsible and equitable AI integration.

CRedit authorship contribution statement

Mazen Abdulwahab Asag: Writing – original draft, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Md Abdullah Al Mamun:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Project administration, Methodology, Conceptualization.

Statements on open data and ethics

Data available on request from the authors. All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional committee. The research design and methodology for this study were approved by the Committee for Advanced Studies and Research (CASR), Islamic University of Technology (IUT), Bangladesh, with reference number: CASR/57/2025/01/Proc/001. Informed consent was obtained from all individual participants included in the study.

Statements on open data and ethics

The study was approved by an ethical committee with ID: CASR/57/2025/01/Proc/001. Informed consent was obtained from all participants, and their privacy rights were strictly observed. The data can be obtained by sending request emails to the corresponding author.

Declaration of generative AI use

During the preparation of this work, the authors used ChatGPT-4 (OpenAI) to improve the readability, grammar, clarity, and refinement of the English language to enhance the overall quality of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the integrity and accuracy of the final published article.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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